

E6680E Wireless Test Set

Testing your Wi-Fi 6E and 802.11be/bn devices

Be Ready for the Future

Maximize your capital investment with Keysight's E6680E Wireless Test Set – engineered for new generation Wi-Fi and 5G device testing. This advanced solution provides you with enough performance and flexibility to address 802.11ax (Wi-Fi 6E), 802.11be (Wi-Fi 7), and 802.11bn (Wi-Fi 8) device test, by delivering up to 800 MHz bandwidth and frequency coverage up to 7.3 GHz.

Choose your bandwidth and frequency range

The IEEE defines 160 MHz bandwidth for Wi-Fi 6 and Wi-Fi 6E, and 320 MHz bandwidth for Wi-Fi 7 and Wi-Fi 8. Chipsets and devices supporting 320 MHz bandwidth are on the rise globally.

The E6680E offers three bandwidths for flexible investment options: 200 MHz, 400 MHz, and 800 MHz. Also offered are two frequency ranges with maximum frequency of 6 GHz or 7.3 GHz. Software licensing enables quick and easy upgrades to bandwidth and frequency later without modifying the E6680E hardware.



Configure Hardware

The E6680E hardware includes the chassis, reference, and embedded controller by default.

Select transceiver, bandwidth, and frequency options

Start by configuring one transceiver (TRX), then add and configure more TRX, if desired. Bandwidth and frequency options apply to TRX 1, TRX 2, and the pair of TRX 3 and TRX 4, as noted.

Option number	Description	Requirement
E6680E-001	TRX 1	Mandatory
E6680E-B2X	Bandwidth, 200 MHz for TRX 1	Default
E6680E-B4X	Bandwidth, 400 MHz for TRX 1	Optional
E6680E-B8X	Bandwidth, 800 MHz for TRX 1	Optional
E6680E-F06	Frequency range, 380 MHz to 6 GHz for TRX 1	Default
E6680E-F07	Frequency range, 380 MHz to 7.3 GHz for TRX 1	Optional
E6680E-002	Add TRX 2	Optional
E6680E-B2X	Bandwidth, 200 MHz for TRX 2	Default
E6680E-B4X	Bandwidth, 400 MHz for TRX 2	Optional
E6680E-B8X	Bandwidth, 800 MHz for TRX 2	Optional
E6680E-F06	Frequency range, 380 MHz to 6 GHz for TRX 2	Default
E6680E-F07	Frequency range, 380 MHz to 7.3 GHz for TRX 2	Optional
E6680E-004	Add TRX 3 and 4	Optional
E6680E-B2X	Bandwidth, 200 MHz for TRX 3 and 4	Default
E6680E-B4X	Bandwidth, 400 MHz for TRX 3 and 4	Optional
E6680E-B8X	Bandwidth, 800 MHz for TRX 3 and 4	Optional
E6680E-F06	Frequency range, 380 MHz to 6 GHz for TRX 3 and 4	Default
E6680E-F07	Frequency range, 380 MHz to 7.3 GHz for TRX 3 and 4	Optional

Select additional hardware options

Option number	Description	Requirement
E6680E-BTS	Downlink measurement capability for all TRX	Optional (required for testing small cells)
E6680E-M44	Up to 4x4 true MIMO in one chassis for all TRX	Optional
E6680E-MTS	Synchronization for up to 8x8 MIMO with two chassis for all TRX	Optional with two E6680E chassis
E6680E-UK6	Commercial calibration certificate with test data for all TRX	Optional

Add IQ streaming capability (optional)

Option number	Description	Requirement	Compatibility
E6680E-STM ¹	IQ streaming In/Out via backplane	Requires M9038A high performance controller and storage modules	Compatible with option 001 and 002
E6680E-CM8	M9038A controller and two PXIe storage modules (total 8TB)	Requires option STM	Not compatible with option CMH
E6680E-CMH	M9038A controller and two PXIe storage modules (total 16TB)	Requires option STM	Not compatible with option CM8

1. E6680E with option STM and CM8 or CMH works with [SJ001A WaveJudge Wireless Analyzer Toolset](#) software for a complete over-the-air PHY and protocol decoding analysis. Contact your nearest Keysight sales representative for more information.

Configure Software

The E6680E wireless test set includes the following software by default.

- Three 50-pack waveform licenses (E6680E-250, 251, 252)
- IQ analysis measurement software (V9060EM0E)
- Sequence analyzer for device applications (V9065EM1E)

Select software applications

Software applications are purchased once for use with all TRX installed in the E6680E chassis. Software applications purchased with new hardware are installed before shipment. Additional software applications can be added later with a software license.

Software applications installed before shipment include a node-locked, perpetual license with 12-month support. To purchase other license types and terms, order the software separately. Then, the end user redeems the licenses and installs the software.

Software applications are based on Keysight's PathWave Signal Generation and Signal Analysis software with modifications to support sequencing and parallel test.

Two types of applications are available. Select one or both types to use with the hardware. V-Apps (e.g., V9080EM0E) provide uplink measurement analysis. Y-Apps (e.g., Y9080EM0E) provide downlink arbitrary waveforms and uplink measurement analysis.

Software application	Description
V9065EM0E	Sequence analyzer BTS applications (requires E6680E-BTS)
V9065EM2E	Parallel analysis of multiple devices
V9071EM0E	GSM/EDGE/Evo measurement
V9073EM0E	W-CDMA/HSPA+ measurement
V9077EM0E	WLAN 802.11a/b/g/j/p/n/af/ah measurement

Software application	Description
V9077EM1E	WLAN 802.11ac/ax measurement
V9077EM2E	WLAN 802.11be and 802.11bn measurement
V9080EM0E	LTE/LTE-A FDD measurement
V9080EM4E	LTE V2X measurement
V9081EM0E	<i>Bluetooth</i> [®] measurement
V9082EM0E	LTE/LTE-A TDD measurement
V9084EM0E	Short-range communications measurement
V9085EM0E	5G NR measurement
V9085EM1E	PAvT measurement (requires V9085EM0E or Y9085EM0E)
V9054EM0E	VMA digital demodulation measurement
V9054EM1E	VMA custom OFDM measurement
Y9071EM0E	GSM/EDGE/Evo waveform and measurement
Y9073EM0E	W-CDMA/HSPA+ waveform and measurement
Y9077EM0E	WLAN 802.11a/b/g/j/p/n/af/ah waveform and measurement
Y9077EM1E	WLAN 802.11ac/ax measurement
Y9077EM2E	WLAN 802.11be and 802.11bn measurement
Y9080EM0E	LTE/LTE-A FDD waveform and measurement
Y9080EM4E	LTE V2X waveform and measurement
Y9081EM0E	<i>Bluetooth</i> [®] waveform and measurement
Y9082EM0E	LTE/LTE-A TDD waveform and measurement
Y9084EM0E	Short-range communications waveform and measurement
Y9085EM0E	5G NR non-signaling waveform and measurement
N7607EMBC ¹	Pathwave signal generation for DFS and DAA profiles, waveform playback

1. Requires E6680E firmware revision E.39 or above, and a valid N7607EMBC license with a version date above 2022.0801

Select other software

Order the following software separately. Then, the end user redeems the licenses and installs the software onto E6680E wireless test set.

Software	Description
ES114000A	Non-signaling RF application
SJ001A	WaveJude Wireless Analyzer Toolset

Add Accessories

Accessory number	Description	Requirements
Y1206A	USB keyboard and optical mouse	Optional
Y1217A	Rack mount and rack rail kit	Optional

Add Services

Hardware support is provided by KeysightCare. Keysight's E6680E Wireless Test Set includes KeysightCare Assured First Year Support (R-55A-001-1) by default. Two-, three-, and five-year extensions are available along with KeysightCare Enhanced upgrades for one, two, three, and five years. For details, visit <http://www.keysight.com/find/keysightcare>.

A one-year, return-to-Keysight warranty (R-51B-001-C) is included by default.

Product assistance is available for solution startup and other custom services. For details, contact your Keysight sales representative or visit <http://www.keysight.com/find/services>.

Configure Upgrades

Add new TRX to an existing chassis to increase capability for testing multiple devices and MIMO. One bandwidth and one frequency option are required for each TRX. New TRX are installed in a Keysight service center.

All other upgrades add capability to existing E6680E test sets with license-only upgrades installed by the end user and no hardware changes.

Option number	Description	Requirements
E6680EU-BTS	Add downlink measurement capability	One license per chassis
E6680EU-BU1	Upgrade bandwidth from 200 to 400 MHz	One license for each TRX
E6680EU-BU2	Upgrade bandwidth from 200 to 800 MHz	
E6680EU-BU3	Upgrade bandwidth from 400 to 800 MHz	
E6680EU-FU1	Upgrade frequency range to 380 MHz to 7.3 GHz	
E6680EU-M44	Add up to 4x4 MIMO in 1 chassis	One license per chassis
E6680EU-MTS	Add synchronization for up to 8x8 MIMO with two chassis	One license for each of two E6680E chassis
E6680EU-TRX	Add one TRX to existing chassis	Hardware upgrade in Keysight service center
E6680EU-B2X	Bandwidth, 200 MHz	Default
E6680EU-B4X	Bandwidth, 400 MHz	Optional
E6680EU-B8X	Bandwidth, 800 MHz	Optional
E6680EU-F06	Frequency range, 380 MHz to 6 GHz	Default

Option number	Description	Requirements	
E6680EU-F07	Frequency range, 380 MHz to 7.3 GHz	Optional	
E6680EU-STM	IQ streaming In/Out via backplane	Requires M9038A high performance controller and storage module.	Compatible with option 001 and 002
E6680EU-CM8	M9038A controller and two PXIe storage modules (total 8TB)	Hardware upgrade in Keysight service center. Requires option STM	Not compatible with option CMH
E6680EU-CMH	M9038A controller and two PXIe storage modules (total 16TB)	Hardware upgrade in Keysight service center. Requires option STM	Not compatible with option CM8

Example Configurations

Following are configurations for some typical test scenarios. Select accessories and services as the final step for all example configurations.

Wi-Fi 6E station (STA) with two transceivers

1. Select hardware.

Option number	Description	Requirement
E6680E-001	TRX 1	Mandatory
E6680E-B2X	Bandwidth, 200 MHz for TRX 1	Default
E6680E-F07	Frequency range, 380 MHz to 7.3 GHz for TRX 1	Optional
E6680E-002	Add TRX 2	Optional
E6680E-B2X	Bandwidth, 200 MHz for TRX 2	Default
E6680E-F07	Frequency range, 380 MHz to 7.3 GHz for TRX 2	Optional

2. Select software.

Software application	Description
Y9077EM0E	WLAN 802.11a/b/g/j/p/n/af/ah waveform and measurement
Y9077EM1E	WLAN 802.11ac/ax waveform and measurement
Y9081EM0E	Bluetooth® waveform and measurement

802.11ax Access Point (AP) with 4x4 MIMO

1. Select hardware.

Option number	Description	Requirement
E6680E-001	TRX 1	Mandatory
E6680E-B2X	Bandwidth, 200 MHz for TRX 1	Default
E6680E-F06	Frequency range, 380 MHz to 6 GHz for TRX 1	Default
E6680E-002	Add TRX 2	Optional
E6680E-B2X	Bandwidth, 200 MHz for TRX 2	Default
E6680E-F06	Frequency range, 380 MHz to 6 GHz for TRX 2	Default
E6680E-004	Add TRX 3 and 4	Optional
E6680E-B2X	Bandwidth, 200 MHz for TRX 3 and 4	Default
E6680E-F06	Frequency range, 380 MHz to 6 GHz for TRX 3 and 4	Default

2. Select additional hardware options.

Option number	Description	Requirement
E6680E-M44	Up to 4x4 true MIMO in one chassis	Optional

3. Select software.

Software application	Description
Y9077EM0E	WLAN 802.11a/b/g/j/p/n/af/ah waveform and measurement
Y9077EM1E	WLAN 802.11ac/ax waveform and measurement
Y9081EM0E	Bluetooth® waveform and measurement

802.11be device verification

1. Select hardware.

Option number	Description	Requirement
E6680E-001	TRX 1	Mandatory
E6680E-B4X	Bandwidth, 400 MHz for TRX 1	Optional
E6680E-F07	Frequency range, 380 MHz to 7.3 GHz for TRX 1	Optional

2. Select software.

Software application	Description
Y9077EM0E	WLAN 802.11a/b/g/j/p/n/af/ah waveform and measurement
Y9077EM1E	WLAN 802.11ac/ax waveform and measurement
Y9077EM2E	WLAN 802.11be and 802.11bn waveform and measurement

WLAN and 5G CPE

1. Select hardware.

Option number	Description	Requirement
E6680E-001	TRX 1	Mandatory
E6680E-B8X	Bandwidth, 800 MHz for TRX 1	Default
E6680E-F07	Frequency range, 380 MHz to 7.3 GHz for TRX 1	Optional
E6680E-002	Add TRX 2	Optional
E6680E-B8X	Bandwidth, 800 MHz for TRX 2	Default
E6680E-F07	Frequency range, 380 MHz to 7.3 GHz for TRX 2	Optional

2. Select software.

Software application	Description
Y9077EM0E	WLAN 802.11a/b/g/j/p/n/af/ah waveform and measurement
Y9077EM1E	WLAN 802.11ac/ax waveform and measurement
Y9085EM0E	5G NR non-signaling waveform and measurement

Wi-Fi 6E small cell

1. Select hardware.

Option number	Description	Requirement
E6680E-001	TRX 1	Mandatory
E6680E-B2X	Bandwidth, 200 MHz for TRX 1	Default
E6680E-F07	Frequency range, 380 MHz to 7.3 GHz for TRX 1	Optional
E6680E-002	Add TRX 2	Optional
E6680E-B2X	Bandwidth, 200 MHz for TRX 2	Default
E6680E-F07	Frequency range, 380 MHz to 7.3 GHz for TRX 2	Optional

2. Select additional hardware options.

Option number	Description	Requirement
E6680E-BTS	Downlink measurement capability for all TRX	Optional

3. Select software.

Software application	Description
Y9077EM0E	WLAN 802.11a/b/g/j/p/n/af/ah waveform and measurement
Y9077EM1E	WLAN 802.11ac/ax waveform and measurement
Y9081EM0E	Bluetooth® waveform and measurement

IQ streaming configuration for WLAN

E6680E supports one-channel or two-channel IQ streaming, with signal bandwidth options of 200, 400, or 800 MHz. Customize the number of channels and bandwidth that best fits your application.

1. Select hardware.

Option number	Description	Requirement
E6680E-001	TRX 1	Mandatory
E6680E-B4X	Bandwidth, 400 MHz for TRX 1	Optional
E6680E-F07	Frequency range, 380 MHz to 7.3 GHz for TRX 1	Optional
E6680E-002	Add TRX 2	Optional
E6680E-B4X	Bandwidth, 400 MHz for TRX 2	Optional
E6680E-F07	Frequency range, 380 MHz to 7.3 GHz for TRX 2	Optional

2. Select additional hardware options.

Option number	Description	Requirement
E6680E-M44	Up to 4x4 true MIMO in one chassis for all TRX	Optional
E6680E-STM	IQ streaming In/Out via backplane	Required
E6680E-CMH	M9038A controller and PXIe storage modules (total 16TB)	Optional

3. Select WaveJudge software.

Software application	Description	
WJ001000A	WaveJudge Wireless Analyzer Application	Required
WJ038100A	WaveJudge WLAN 802.11 a/b/g/n/ac	Required
WJ038101A	WaveJudge WLAN 802.11ax	Optional
WJ038102A	WaveJudge WLAN 802.11be	Optional
WJ0380M0A	WaveJudge WLAN 4096QAM	Optional

Upgrade from 802.11ax to 802.11be/bn

1. Select hardware upgrades (delivered with software-only licenses).

Option number	Description	Requirement
E6680EU-BU1	Upgrade bandwidth from 200 to 400 MHz	One license for each TRX
E6680EU-FU1	Upgrade frequency range to 380 MHz to 7.3 GHz	One license for each

2. Select customer installation of license certificates.
3. Add software separately. Select desired license details, certificate delivery, and application.

Software application	Description
Y9077EM2E	WLAN 802.11be and 802.11bn waveform and measurement

Add 4x4 MIMO to existing 2x2 MIMO for Wi-Fi 6E AP

1. Select hardware upgrades.

Option number	Description	Requirements
E6680EU-TRX	Add one TRX to existing chassis	Hardware upgrade in Keysight service center
E6680EU-B2X	Bandwidth, 200 MHz	Default
E6680EU-F07	Frequency range, 380 MHz to 7.3 GHz	Optional
E6680EU-TRX	Add one TRX to existing chassis	Hardware upgrade in Keysight service center
E6680EU-B2X	Bandwidth, 200 MHz	Default
E6680EU-F07	Frequency range, 380 MHz to 7.3 GHz	Optional

Add 802.11be 8x8 MIMO to existing 802.11ax AP

1. Select hardware upgrades (delivered with software-only licenses).

Option number	Description	Requirement
E6680EU-BU1	Upgrade bandwidth from 200 to 400 MHz	One license for each TRX
E6680EU-FU1	Upgrade frequency range to 380 MHz to 7.3 GHz	One license for each
E6680EU-M44	Add up to 4x4 MIMO in one chassis	One license for each E6680E chassis
E6680EU-MTS	Add synchronization for up to 8x8 MIMO with two chassis	One license for each of two E6680E chassis

2. Select customer installation of license certificates.
3. Add software separately. Select desired license details, certificate delivery, and application.

Software application	Description
Y9077EM2E	WLAN 802.11be and 802.11bn waveform and measurement

Add 802.11be small cell to existing 802.11ax device verification configuration

1. Select hardware upgrades (delivered with software-only licenses).

Option number	Description	Requirement
E6680EU-BTS	Add downlink measurement capability for all TRX	
E6680EU-BU1	Upgrade bandwidth from 200 to 400 MHz	One license for each TRX
E6680EU-FU1	Upgrade frequency range to 380 MHz to 7.3 GHz	One license for each

2. Select customer installation of license certificates.
3. Add software separately. Select desired license details, certificate delivery, and application.

Software application	Description
Y9077EM2E	WLAN 802.11be and 802.11bn waveform and measurement

The E6680E is the solution to test Wi-Fi 6E, and 802.11be/bn devices

Speed your test development and improve test time of Wi-Fi 6E, and 802.11be/bn devices in production with just the right capability and flexibility.

- Choose bandwidth and frequency range based on your needs today and be confident that the hardware will support your needs tomorrow with software-only upgrades.
- Get the flexibility you need to optimize your test cases in manufacturing with up to 4 transceivers and 16 ports.
- Simplify your production test system configuration with internal switching between ports.
- Verify Wi-Fi devices across the workflow with E6680A and E6680E solutions from Keysight

An optimal solution for your test bench or production line, the E6680E is ready to test your WLAN devices. Right out of the box, its familiar application software is licensed to your specifications. Software-only upgrades, three bandwidth options, and two frequency options ensure you get just what you need today and tomorrow.

Visit www.keysight.com/find/e6680e, or contact your sales representative for details.

Visit www.keysight.com/find/e6680a for more details about Keysight's E6680A Wireless Test Set.

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